



OPEN Optimized segmentation of overlapping cervical cells based on Mask2Former and denoising

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Cell instances segmentation in cervical cytology images have crucial importance for cancer screening and automatic biomedical image analysis. However, that the highly overlapping cells cause ambiguous boundaries makes accurate cell segmentation still a challenging task. This paper proposes a novel transformer-based neural network to segment overlapping cervical cells, by using the technique of denoising with Mask2Former. The model optimization includes two aspects. One is that class embeddings of ground truth categories are used as extra content queries to transformer decoder. The other is that noised ground truth masks are fed into each layer of transformer decoder and then the model is trained to reconstruct the original masks. Compared with the best result on ISBI2014 dataset, the proposed model obtains performance improvement by 3.4% on DSC, 2.3% on TPRp and 2% on FNRo, respectively. Especially, the masks predicted by our model are more precise than other methods and our previous models. Those results show that the proposed method could be effective in segmenting overlapping cells and help pathologists detect cell lesions.

Keywords Mask2Former, Denoising, Overlapping cervical cells, Noised masks, ISBI

Cervical cancer is one of the most common cancers among women worldwide and is the second leading cause of cancer death¹. It was reported that over 340 thousand women died of cervical cancer². Early detection is essential for the cancer to be treated duly. Cytology examination is one of the most important screening techniques for cervical cell lesions, including Pap Smear and liquid-based cytology. Digital images are usually obtained from those tests. Generally, pathological diagnosis of cervical cancer mainly depends on experts observing the deformation of cells in the images. Liquid-based cytology, such as Thinprep Cytological Test (TCT), is more advanced than Pap Smear. Liquid-based cytology can present high-quality images of cell samples, which are essential for visual inspection. However, it is time-consuming and inefficient for manually detecting abnormal cells in cytology images, due to the demand for extensive human efforts. Therefore, computer-aided systems are crucial for improving the efficiency and accuracy of diagnosis. Cellpose³ is a deep learning tool for biomedical image analysis, primarily used for cell segmentation. Cellpose achieves impressive cell segmentation performance, making it suitable for a wide range of biomedical image analysis tasks. But, the models used by Cellpose are not capable of dealing with the highly overlapped cells in cervical cytology images. However, Cellpose represents a highly commendable attempt to apply deep learning methods in clinical practice. Various screening techniques have been widely used for detecting cervical cancers nowadays, but this process is likely to be affected by many factors such as fatigue eyes and blurred edges of cells, causing misdiagnosis. Accurate cell segmentation could overcome those potential factors leading to misdiagnosis. Therefore, it's extremely necessary to develop more effective and economical methods for detecting cervical cell edges.

In recent years, deep learning has played a very important role in various vision tasks, including image classification, semantic segmentation and instance segmentation. Many deep-learning based methods have also been proposed to deal with various biomedical images, including two-dimensional images (such as in work⁴) and one-dimensional images (such as in work⁵). As for cervical cytology images, the detection of nuclei and cytoplasm is the core task of cervical cell pathological examination. Cytoplasmic characteristics have been shown to be critical for discriminating abnormal cells. Once the precise segmentation of cytoplasm is obtained, quantitative metrics such as diameters and rates of deformation can be calculated. Based on this information,

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the degrees of cervical cell lesions can be inferred automatically. Especially, accurate segmentation of cytoplasm from cell clumps of high density in real images from clinical tests can clearly present the shape and volume characteristics of cervical cells, which could tremendously improve the effectiveness of conducting cervical cancer screening.

However, actual cervical cytology images often contain mucus, blood or other debris, despite advanced smearing techniques (such as TCT) being used. And the cells in the images often overlap to the extent that cytoplasm boundaries in cell clumps are likely to be visually indistinguishable. Those facts make it still a challenging task to precisely segment every single cell in the cervical cytology images. Thus, scholars around the world have proposed many methods to address the problem of cervical cell segmentation recently.

On the other hand, deep learning based methods demand numerous samples and corresponding annotations for training. But manual annotations of all cells in images are very tedious and time-consuming. Thus, biomedical datasets usually have only hundreds or thousands of image samples, which are very precious. The IEEE International Biomedical Imaging Society held the first and second cervical cell image segmentation challenges in 2014 and 2015, and published two high quality datasets containing cervical cytology images and manual annotations. Those two datasets have motivated the study of overlapping cells segmentation tremendously. In 2022, Liu et al.¹² developed a new dataset of cervical cytology images named Cx22, which consisted of the completely annotated labels of the cellular instances (including both nuclei and cytoplasm instances). The Cx22 dataset has similar quality to the ISBI datasets, but they are stained with different techniques. Although the Cx22 dataset increases the number of cervical cytology images for training and testing, the total number is somehow insufficient. Therefore, the weights pre-trained on large scale datasets (such as ImageNet¹¹ and COCO datasets⁹) are used to initialize our model.

This research introduces a new transformer-based model by enhancing the original Mask2Former, to address the problem of segmenting overlapped cells in cervical cytology images. The main contributions of our work are:

- We propose a new model to segment cytoplasm from dense cell clumps in cervical cytology images. The model is created by adding noised ground truth masks and extra class embeddings to the original Mask2Former, to improve the segmentation performance.
- Both the classical ISBI datasets and the new Cx22 dataset are used to test the segmentation performance of the proposed model.
- Experiments show that the proposed method achieves state-of-the-art cytoplasm segmentation results on all four metrics on both the ISBI2014 and Cx22 datasets

The rest of this paper is organized as follows. Section “Related works” briefly discusses previous papers about the cell segmentation in cervical cytology images. Section “Methods” presents detailed information about the datasets, the structure of the proposed model and implementation parameters. Section “Results” gives the main results, quantitative comparison between related works, and visualization of segmentation results. A comprehensive discussion about the pros and cons of the proposed model is presented in Section “Discussion”. Finally, the conclusion and future works to improve the segmentation performance are given in Section “Conclusion and future works”.

Related works

Accurate cervical cell segmentation is a very challenging task even for human experts, and also plays a very important role in the computer-aided system for automatic diagnosis. Generally speaking, the most intricate technical issue is segmenting every single cell from dense clumps in cervical cytology images, where cells overlap at a high rate. In the process of cell sample preparation, uneven illumination and staining in dye concentration would cause low contrast between different components in the cytology image. Blood cells, bacteria and mucus would also increase the difficulty of overlapping cell segmentation⁶. In order to promote the research of overlapping cervical cell segmentation, the IEEE International Biomedical Imaging Society held the first and second cervical cell image segmentation challenges in 2014 and 2015, during which two high-quality datasets of cervical cytology images and their annotations were made publicly available. Those datasets make it possible to compare the performance of different cell segmentation methods. The new published dataset Cx22 augments the dataset of cervical cytology images.

Ushizima et al.²⁰ used graph-based and global search truncation algorithm to obtain cell clumps based on the similarity of the intensity of adjacent pixels. They divided the overlapping cells by polygons, which were not smooth enough. Phoulady et al.²² proposed a method based on iterative thresholds and the regular level set algorithm to segment cells. They segmented the cell clumps and nucleus areas by iterative thresholds. Then smooth boundaries of cytoplasm were obtained by applying the regular level set having the shape of ellipse. To further improve the performance of cytoplasm segmentation, Phoulady et al.²³ defined a weight vector to detect candidate edge pixels, and then utilized a smoothing filter to smooth the edges. Lee et al.²⁴ used superpixel generation and triangle thresholding to detect cell masses. Cell cytoplasm was segmented by superpixel partitioning and then improved by cell-wise contour refinement with graph cuts. Wang et al.²⁵ generated the weight graph by candidate contour points and utilized the dynamic programming algorithm to find the shortest path, where the shortest path was the coarse segmentation contour. The level set model was also used to smooth the edges. Tareef et al.²⁶ introduced a multi-pass fast watershed-based method to segment both nuclei and cytoplasm from large cell masses of overlapping cervical cells. The method contained three watershed passes, to obtain a better representation of cell shapes. They tested their method on both ISBI2014 and ISBI2015 dataset, and achieved superior segmentation performance at that time. Nuclei localization were obtained in the first-pass. Then the technique of watershed was used again to get cell segmentation in the second-pass. The third

watershed pass was used to extract the individual cytoplasm of cluster that were not segmented in previous steps due to the obscure boundaries of the cells.

In recent years, neural network methods have achieved outstanding results in the field of medical image classification and segmentation. Although, these methods are computationally expensive, they outperform other methods in many vision tasks. Most of those approaches are based on the architectures of convolutional neural network (CNN) and transformer, as well as the hybrid model of CNN and transformer. Tareef et al.²⁷ combined the deep learning model and dynamic shape model to separate individual nucleus and cytoplasm. This method solved the problem of low sensitivity of cervical cell segmentation to some extent. Wan et al.²⁸ introduced a double-window based cellular detection model to detect every single cell, where TerausNet was used to classify the pixels into three categories (i.e. background, nucleus, cytoplasm). Then a modified DeepLab model was used to refine the edges of cytoplasm and nuclei. But instead of the original images (1024×1024) in ISBI2015 dataset, cropped images of size 224×224 were utilized to test the segmentation performance of their method. Zhang et al.²⁹ addressed a new framework based on Attention U-Net network and graph-based random walk to extract both nucleus and cytoplasm from clumps with overlapping cervical cells. Their experiments showed that random walk worked very well on feature maps generated by Attention U-Net, to obtain the segmentation of every individual cell from dense cell regions. Song et al.³⁰ proposed a learning-based model with robust shape priors to segment individual cell, where multi-scale convolutional networks were adopted to learn features on various levels. They also incorporated high-level shape information to guide segmentation, where cell boundaries were blurred due to cells overlapping. Mahyari et al.³¹ used a convolutional network to obtain feature maps, with which the approximate location of nuclei and cytoplasm could be identified. Then multi-layer random walk image segmentation was used with cell nuclei as hard initial seeds, to estimate cytoplasm segmentation. Lastly, both cytoplasm segmentation were combined to refine the final cell segmentation. Zhao et al.³² proposed a novel star-convex polygon-based convolutional neural network with an encoder-decoder structure, to address the segmentation of adherent cervical cells. A polygon-based adaptive non-maximal suppression algorithm was used to screen the polygon proposals, aiming to achieve accurate segmentation of adherent cells. Jiang et al.³³ proposed a De-overlapping network in a decompose-and-recombined strategy. Firstly, a dual-path region segmentation module was introduced to decompose the cell clusters into intersection and complement regions. Secondly, a semantic consistency guided recombination module was utilized to integrate the two regions. Especially, they designed a mask-guided region proposal strategy to refine inner-cell instance prediction. They tested the cell segmentation performance on the ISBI2014 and CPS datasets. Jiang et al.³⁴ introduced a dual-stream self-attention to enhance the learning of lesion-specific features. They aimed to focus more on lesion cells and cell nuclei of diagnostic significance. Our previous work³⁸ used PointRend module with baseline model Mask-RCNN to refine the edge pixels of predicted cytoplasm. Our another previous work³⁹ built an enhanced mask transformer based on DETR to segment cytoplasm in cervical cytology images.

Method
Datasets

In 2014 and 2015, the first and second Overlapping Cervical Cytology Image Segmentation challenges (ISBI 2014 and ISBI 2015⁷) were held, with two high-quality datasets of cervical cytology images and their ground-truth segmentation made publicly available. The first dataset (from ISBI 2014) consists of 135 synthetic (45 images for training and 90 images for validating) and 8 real cervical cytology extended depths of field (abbreviated as EDF) images in the training set and 810 synthetic and 8 real cervical cytology EDF images in the testing set. The synthetic images are generated by affine transformations of isolated cells in real EDF images. It should be perceived that the real EDF images in this dataset were released without annotations of individual cytoplasm. So they were not used in our paper. The second dataset (from ISBI 2015) contains 8 real cervical cytology EDF images in the training set and 9 real ones in the testing set, with both ground-truth annotations of cytoplasm. In 2022, Liu et al.¹² introduced a new dataset of cytology images named Cx22, which was based on the CCEDD dataset⁸. They cropped the original images to 512×512 pixels and then manually delineated the contours of 7473 instances of nucleus and cytoplasm across 1320 images. It should be noticed that some images in Cx22 are cropped from the original image with a rectangle at first and then padded to the size of 512×512 by white pixels. The Cx22 dataset contains 400 images in the training dataset and 100 ones in the testing dataset. As for the main difference between the datasets ISBI2014 and ISBI2015, an image from ISBI2014 has size of 512×512 with 2-10 cells in it while an image from ISBI2015 has size of 1024×1024 with more than 40 highly overlapped cells. The images from Cx22 are similar to images from the ISBI datasets, but they are stained with different smearing techniques. Table 1 gives out the detailed information of the training and testing datasets used in this paper. There are 1263 cervical cytology image samples and 199 ones in the training and testing datasets respectively. Concretely, the training dataset used in this paper consists of 855 synthetic cell images from ISBI2014 and 8 EDF images from the training set of ISBI2015 and 400 images from Cx22 training dataset. The testing dataset is composed of 90 synthetic images for validating from ISBI2014 and 9 EDF images from testing set of ISBI2015 and 100 images from Cx22 testing dataset. Data augmentation can not only improve generalization ability of the

	Total	ISBI2014	ISBI2015	Cx22
Training set	1263	855	8	400
Testing set	199	90	9	100

Table 1. The number of cell images in three datasets and their partitioning.

model, but also reduce the probability of overfitting. So data augmentation techniques such as vertical flip are used in the training process of our model.

Three typical samples of cytology images and their cytoplasm annotations are presented in Fig. 1. Figure 1b shows that cells highly overlap at the bottom left of the image, to make the edges of cells very blurred. As can be seen from Fig. 1c, the image is padded with a white rectangle at the bottom, and contains scattered cell debris.

Structure of the model

Although neural networks based on convolution have made great success in vision tasks, those methods often employ many hand-crafted components (such as non-maximum suppression). The transformer based neural network DETR¹³ proposed by Carion et al. in 2020 was the first fully end-to-end object detector. The DETR model has been proven to be very effective in various vision tasks, including semantic segmentation and instance segmentation. Since then, many models based on DETR have been proposed for object detection and achieved very impressive results (such as works^{14–16}). Another typical transformer-based model is Swin-Transformer¹⁷, which often serves as a backbone network to extract multi-scale features. Li et al.¹⁸ presented a thorough discussion about transformer-based visual segmentation. In a word, transformer-based networks have great potential in various vision tasks, including the segmentation and classification of biomedical images.

The original Mask2Former¹⁹ proposed by Cheng et al. is an architecture capable of addressing general image segmentation tasks (panoptic, instance and semantic segmentation). They proposed a new masked attention module to extract localized features, instead of standard cross attention. Mask2former achieved a very impressive instance segmentation result with 50.1 AP on COCO dataset, which set a new state-of-the-art result at that time. Based on Mask2former, we not only improve performance of baseline model but also accelerate its convergence by adding noised ground truth masks to the input of each layer in the transformer decoder. The overall framework of our proposed model is depicted in Fig. 2.

As can be seen from Fig. 2, our proposed model is built on the original model of mask2former, by adding class embeddings and noised ground truth masks. In detail, the traditional convolutional neural networks such as ResNet50 or the Swin transformers of different size are often used as the backbone network, to extract multi-scale features from the original images. Assuming that the Swin_B network is used as the backbone network, four features (namely {'res2', 'res3', 'res4', 'res5'}) of different sizes are extracted and then fed into the pixel decoder. Deformable encoder with six layers takes the multi-scale features {'res2', 'res3', 'res4'} added with positional embeddings as input, to get O_1, O_2, O_3 . The feature maps O_1, O_2, O_3 are interpolated to the same size as 'res5'. One convolutional layer with 1×1 filter is used to linear map 'res5'. Then the interpolated features and the linear mapped results are added to get O_4 , which would be utilized to generate the mask predictions. One multi-scale feature from pixel decoder is fed into one transformer decoder layer at a time. An extra set of queries and masks (the yellow components in Fig. 2) are fed into the transformer decoder. Class embeddings of ground truth categories are used as the content queries. It should be noticed that the class embeddings are only used in the first layer of transformer decoder. Ground truth masks may be too easy for the model to predict. So we further propose to use noised ground truth masks. Those noised ground truth masks are fed into the

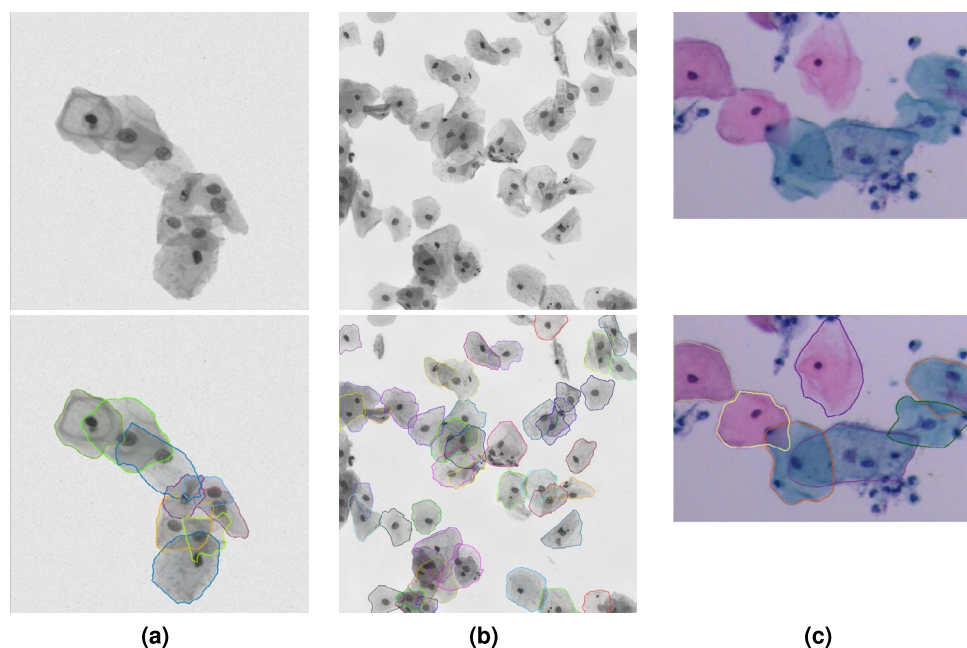


Figure 1. Typical cytology images (first row) with their ground-truth cytoplasm masks (second row) in the training dataset: (1a) Synthetic cell image with low overlapping rate (from ISBI2014); (1b) Real EDF image with high overlapping rate (from ISBI2015); (1c) Cell image with moderate overlapping rate (from Cx22).

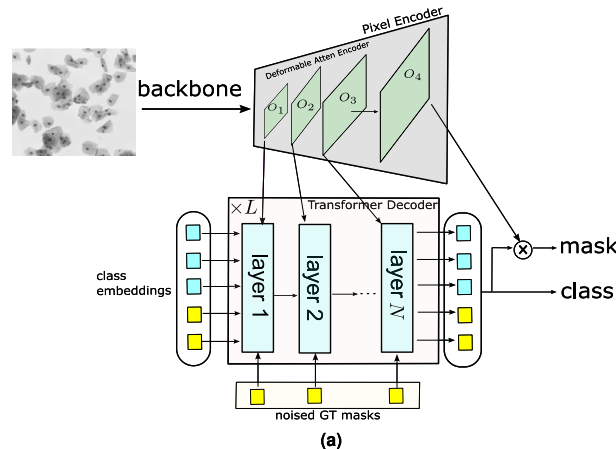


Figure 2. The proposed framework of mask2former with extra class embeddings and noised ground truth masks (which are concatenated with the output of each layer in transformer decoder).

transformer decoder. Then the model is trained to reconstruct the original ones. The predictions of noised masks are assigned, rather than matched by Hungarian Algorithm, with the ground truth masks. Because the noised masks are relatively easy for the model to denoise, this technique would accelerate the model convergence and improve the segmentation performance to some extent. It should be noticed that those two techniques are only used in the process of training. Therefore, the model is exactly the same as mask2former during inference.

Implementation details

In this paper, the traditional CNN ResNet50 and Swin transformers are used as the backbone networks to extract multi-scale features. Swin transformers of different sizes (i.e. Swin Tiny, Swin Base and Swin Large) are used for testing in this paper. Deep learning models often need lots of images for training, whereas cervical cytology image datasets often contain hundreds or thousands of images. Therefore, the weights pretrained on the famous COCO datasets and ImageNet with twenty-one thousand categories are used to initialize the model. Then the model is further trained and fine-tuned on the ISBI and Cx22 datasets. As for the pixel decoder, six multi-scale deformable attention transformer modules are applied to the feature maps with resolution $\frac{1}{8}$, $\frac{1}{16}$ and $\frac{1}{32}$, which are extracted from input images by the backbone network. An upsampling layer is applied on the feature map of resolution $\frac{1}{8}$ to generate the feature map of resolution $\frac{1}{4}$, which is used with the output of transformer decoder to predict the final masks. The transformer decoder has three modules with 100 queries, while each module has three layers (i.e. $L = 3$, $N = 3$ in Fig. 2). Also 100 ground truth masks with noised pixels are used to accelerate the convergence of the model. The ratio of noised pixels to the total pixels in the ground truth mask is $\lambda_p = 0.2$. The number of noised points M is calculated by equation: $M = \lambda_p \times \sum([p_{i,j}])$, where $p_{i,j} = 0$ or 1. Namely, 20 percent of the pixels in a ground truth mask are randomly selected and flipped (i.e. 0 is flipped to 1 and vice versa). The loss of our model consists of cross-entropy loss (L_{cls}), dice loss (L_{dice}) and mask loss (L_{mask}), where the coefficients are 2, 5 and 5 respectively. Namely, $Loss = 2 \times L_{cls} + 5 \times L_{dice} + 5 \times L_{mask}$. The denoising class loss (L_{cls_dn}), denoising dice loss (L_{dice_dn}) and denoising mask loss (L_{mask_dn}) with coefficients 2, 5, 5 respectively, are included in the total loss while training, but not while testing. The optimizer is AdamW with momentum 0.9. The implementation of our proposed model is based on Mask2former and the detection platform Detectron2. The pretrained weights are from Mask2Former model zoo. Both the baseline model Mask2Former and our proposed model were trained for about 300 epochs. The experiments were carried out on a desktop platform with two GPUs (Nvidia RTX 3090 and RTX 4090) running Debian 11.

Performance measures

In order to compare the segmentation performance with the state-of-the-art methods, we adopt four widely used evaluation metrics: pixel-based dice similarity coefficient (DSC), pixel-based true positive rate (TPRp), pixel-based false positive rate (FPRp) and object-based false negative rate (FNRo). The dice similarity coefficient (abbreviated as DSC) of two regions A and B is defined as:

$$DSC = 2 \times \frac{|A \cap B|}{|A| + |B|}$$

where operator $|\cdot|$ means the area of a region. True positive rate and false positive rate are usual evaluation metrics. Their definitions are as followed: $TPRp = TP/(TP + FN)$, $FPRp = FP/(FP + TN)$. The DSC threshold of 0.7 is recommended by the two ISBI challenges and thereafter is widely used in many works. We adopt the following values of DSC threshold: {0.7, 0.8} in this paper for comparison. A cell in an image is not counted in the calculation of TPRp and FPRp if there is no predicted segmentation that has a DSC greater than

the specific threshold with it. The ratio of cells of this kind to total cells in an image is FNRo. In summary, a higher value of TPRp with a lower value of FPRp means better cell segmentation. Lower value of FNRo indicates that more cells are successfully predicted by the proposed model. The standard metrics for instance segmentation are also reported, including mAP, AP50 and AP75.

Results

Evaluation and comparison

The original transformer-based model Mask2Former is a widely used model for semantic, panoptic and instance segmentation. It also has achieved very impressive results in those three vision tasks on the widely used datasets, such as the COCO and ADE20K¹⁰ datasets. To further accelerate the convergence and improve the performance of the model, we optimize the Mask2Former model by adding noised ground truth masks and then fine-tune the proposed model on the cervical cytology datasets.

As shown in Fig. 3a, the baseline model Mask2Former starts converging at 300 epochs. And our proposed model starts converging at about 200 epochs. Figure 3b shows the similar characteristics as Fig. 3a.

The instance segmentation method proposed in this paper aims to classify each pixel in cervical images into two categories: background or cytoplasm. To validate the effectiveness of our method, we compare the segmentation performance with other state-of-the-art methods on three datasets, including ISBI2014, ISBI2015 and Cx22. The detailed results of our model are presented in Tables 2, 3 and 4. It should be noted that all the four metrics are reported on three datasets respectively. And all the floats are rounded to the nearest three decimal places. Specifically, the baseline model Mask2Former (abbreviated as Mask2 in the following Tables) and the proposed model are trained and tested with four different backbone networks, namely ResNet50, Swin_Tiny, Swin_Base and Swin_Large. As can be seen from Tables 2, 3 and 4, the baseline model Mask2Former and our method achieve the approximate performance on DSC, TPRp and FPRp. However, the proposed method obtains performance improvement on FNRo by 2%, 2.87% and 1% on datasets ISBI2014, ISBI2015 and Cx22 respectively (the averaged value of FNRo across four backbones). Lower FNRo means more cells are segmented

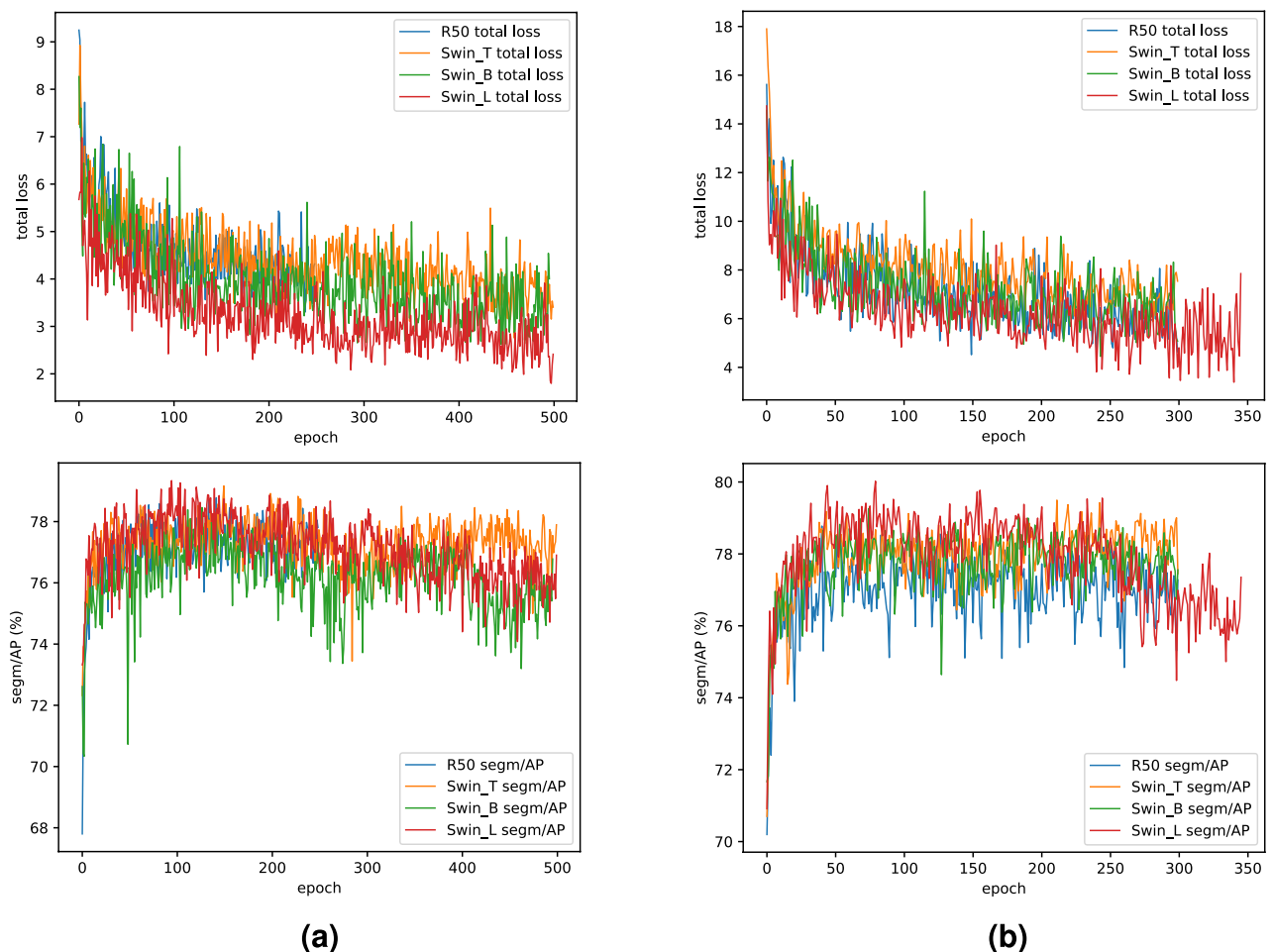


Figure 3. (a) The total losses and average precision of Mask2Former; (b) the total loss and average precision of our proposed model.

Methods	DSC	TPRp	FPRp	FNRo
Mask2 (R50)	0.971 (0.971)	0.971 (0.971)	0.0008 (0.0007)	0.003 (0.007)
Mask2(Swin_T)	0.972 (0.973)	0.971 (0.971)	0.0008 (0.0006)	0.003 (0.007)
Mask2(Swin_B)	0.972 (0.973)	0.973 (0.973)	0.0007 (0.0007)	0.002 (0.007)
Mask2(Swin_L)	0.973 (0.974)	0.969 (0.969)	0.0006 (0.0005)	0.002 (0.008)
Ours (R50)	0.971 (0.973)	0.967 (0.973)	0.0006 (0.0005)	0.001 (0.010)
Ours (Swin_T)	0.973 (0.974)	0.969 (0.969)	0.0005 (0.0005)	0.001 (0.007)
Ours (Swin_B)	0.974 (0.975)	0.973 (0.974)	0.0006 (0.0006)	0.0 (0.007)
Ours (Swin_L)	0.974 (0.975)	0.972 (0.973)	0.0006 (0.0006)	0.0 (0.007)

Table 2. Segmentation performance with DSC threshold=0.7 on the ISBI2014 dataset. The floats in the bracket are the metrics with DSC threshold = 0.8.

Methods	DSC	TPRp	FPRp	FNRo
Mask2 (R50)	0.890 (0.916)	0.932 (0.937)	0.001 (0.0009)	0.261 (0.379)
Mask2(Swin_T)	0.889 (0.919)	0.917 (0.936)	0.001 (0.0008)	0.223 (0.363)
Mask2(Swin_B)	0.895 (0.911)	0.896 (0.911)	0.001 (0.0006)	0.269 (0.413)
Mask2(Swin_L)	0.889 (0.920)	0.905 (0.920)	0.001 (0.0006)	0.266 (0.399)
Ours (R50)	0.887 (0.917)	0.923 (0.931)	0.001 (0.0008)	0.233 (0.373)
Ours (Swin_T)	0.892 (0.917)	0.910 (0.923)	0.001 (0.0007)	0.239 (0.347)
Ours (Swin_B)	0.881 (0.911)	0.910 (0.925)	0.001 (0.0009)	0.202 (0.349)
Ours (Swin_L)	0.888 (0.921)	0.914 (0.929)	0.001 (0.0007)	0.230 (0.380)

Table 3. Segmentation performance with DSC threshold=0.7 on the ISBI2015 dataset. The floats in the bracket are the metrics with DSC threshold = 0.8.

Methods	DSC	TPRp	FPRp	FNRo
Mask2 (R50)	0.954 (0.956)	0.949 (0.950)	0.0008 (0.0008)	0.008 (0.019)
Mask2 (Swin_T)	0.953 (0.955)	0.945 (0.947)	0.0008 (0.0007)	0.006 (0.018)
Mask2 (Swin_B)	0.951 (0.953)	0.945 (0.946)	0.0009 (0.0008)	0.010 (0.023)
Mask2 (Swin_L)	0.952 (0.955)	0.945 (0.947)	0.0009 (0.0008)	0.009 (0.025)
Ours (R50)	0.950 (0.952)	0.942 (0.943)	0.0009 (0.0008)	0.007 (0.016)
Ours (Swin_T)	0.950 (0.952)	0.935 (0.937)	0.0007 (0.0006)	0.009 (0.022)
Ours (Swin_B)	0.951 (0.953)	0.942 (0.944)	0.0008 (0.0008)	0.007 (0.018)
Ours (Swin_L)	0.951 (0.954)	0.944 (0.946)	0.0009 (0.0008)	0.006 (0.019)

Table 4. Segmentation performance with DSC threshold=0.7 on the Cx22 dataset. The floats in the bracket are the metrics with DSC threshold = 0.8.

successfully by the model. For the whole testing dataset, the standard instance segmentation metrics AP, AP50 and AP75 are also calculated. The baseline model Mask2Former with Swin_B backbone achieves a 76.68 mAP, 92.33 AP50 and 83.97 AP75. The proposed model with Swin_B backbone in this paper achieves a 77.56 mAP, 93.94 AP50 and 85.02 AP75. According the Tables 2 and 3, our proposed model which uses Swin_B as the backbone network achieves the best result with 0.202 FNRo on the ISBI2015 dataset under DSC threshold = 0.8. As the cells in the images from ISBI2015 dataset overlap at the highest rates, segmenting cells from those images is the most challenging task. Therefore, the proposed model with Swin_B as the backbone network would be selected for comparison with other state-of-the-art methods.

In the following, we focus on the comparative analysis of various methods for cervical cytology image segmentation quantitatively and qualitatively. Not only results based on traditional methods but also results of modern techniques are used for comparison. The performance of our proposed method and other impressive segmentation methods are compared in Tables 5, 6 and 7.

On the ISBI2014 dataset, most of the models listed in Table 5 perform very well for cytoplasm segmentation despite the existence of stacking cells. Specifically, Lu et al.⁷ achieved very impressive segmentation results with DSC 0.94, TPRp 0.95 on the ISBI2014 dataset in 2015. Table 5 also shows that both the Mask2Former and our proposed model outperform other methods on all four metrics by a moderate margin. Concretely, our proposed model outperforms state-of-the-art result by 3.4% on DSC, 2.3% on TPRp and 2% on FNRo respectively. The metric FNRo of our model is 0, which means every cell in each image from ISBI2014 is predicted with a mask

Methods	DSC	TPRp	FPRp	FNRo
Lu ⁷	0.94±0.07	0.95±0.07	0.004±0.01	0.02±0.06
Wang ²⁵	0.90±NA	0.93±NA	NA	0.068±NA
Tareef ²⁶	0.90±0.074	0.95±0.072	0.004±0.004	0.11±0.167
Wan ²⁸	0.93±0.04	0.93±0.05	0.001±0.002	0.11±0.13
Zhang ²⁹	0.93±0.07	0.94±0.06	0.003±0.004	0.05±0.13
Mahyari ³¹	0.89±0.07	0.90±0.09	0.004±0.003	0.10±0.09
Zhao ³²	0.926±NA	0.867±NA	0.002±NA	0.055±NA
Jiang ³³	0.921±NA	0.948±NA	NA	0.162±NA
Mask2 (Swin_B)	0.972±0.018	0.973±0.017	0.0007±0.0005	0.0024±0.016
Ours (Swin_B)	0.974±0.017	0.973±0.019	0.0006±0.0005	0.0±0.0

Table 5. The quantitative comparison results on the ISBI 2014 dataset with DSC threshold=0.7.

Methods	DSC	TPRp	FPRp	FNRo
Phoulday ²³	0.87±NA	0.88±NA	NA	0.21±NA
Lee ²⁴	0.88±0.09	0.88±0.12	0.001±0.001	0.43±0.17
Wang ²⁵	0.88±NA	0.85±NA	NA	0.32±NA
Tareef ²⁶	0.85±0.079	0.94±0.08	0.002±0.002	0.34±0.13
Wan ²⁸	0.92±0.05	0.91±0.05	0.001±0.003	0.24±0.19
Song ³⁰	0.89±NA	0.92±NA	0.002±NA	0.26±NA
Mask2 (Swin_B)	0.896±0.023	0.896±0.042	0.001±0.0002	0.269±0.132
Ours (Swin_B)	0.881±0.027	0.910±0.042	0.001±0.0003	0.202±0.110

Table 6. The quantitative comparison results on the ISBI 2015 dataset with DSC threshold = 0.7. *The test dataset consists of 210 images of size 224×224 extracted from real EDF images.

Methods	DSC	TPRp	FPRp	FNRo
Liu ^{*12}	0.935±0.041	0.914±0.062	0.0008±0.0015	0.026±0.045
Mask2 (Swin_B)	0.951±0.013	0.945±0.023	0.0009±0.0008	0.010±0.028
Ours (Swin_B)	0.951±0.014	0.943±0.025	0.0008±0.0008	0.007±0.028

Table 7. The quantitative comparison results on the Cx22 dataset with DSC threshold = 0.7. *The results with the highest value of DSC.

having DSC above the threshold (i.e. 0.7). There is little segmentation performance difference between the Mask2Former and our model on other metrics.

On the ISBI2015 dataset, our model doesn't outperform other methods on four metrics. For example, the DSC of our model is a little higher than that of Tareef²⁶, but the metric TPRp is a bit lower than Tareef's result. The averaged metrics among six compared methods in Table 6 are DSC=0.8816, TPRp=0.8966, FPRp=0.0015 and FNRo=0.3. The performance of the proposed model is above the averaged results. It's worth mentioning that the FNRo=20.2% is fairly lower than the averaged FNRo=30% by almost 10 percent. However, none of the publicly available works has achieved superior results on all four metrics, to the best of our knowledge. Some recent survey papers, such as the works of Fang et al.³⁵, Chen et al.³⁶ and Huang et al.³⁷, present comprehensive reviews about the advances of cervical cytology image segmentation.

Few works about cytoplasm segmentation on the Cx22 dataset have been published, due to the fact that the Cx22 dataset was released in 2022. Therefore, Table 7 only contains the results presented by the original authors of Cx22 dataset. As can be seen from this table, both Mask2Former and our model outperform the results of Liu et al.¹² on all four metrics.

In summary, our proposed method in this paper outperforms state-of-the-art cervical cytology image segmentation methods on the ISBI2014 dataset, and is a little better than the averaged values on the ISBI2015 dataset. Because the Cx22 dataset has not been widely used to verify the performance of segmenting cytoplasm, the comparison is limited. On the other hand, the technique of denoising makes the proposed model converge quickly only after a small number of epochs.

Visualization of segmentation results

We present the segmentation results of the baseline model Mask2Former and our proposed model on cervical cytology images with overlapped cells by visualization. In Fig. 4, three typical images are selected from the testing dataset, to show the ground truth cytoplasm annotations and the masks predicted by our model.

As can be seen from Fig. 4, both the Mask2Former and our model can effectively segment almost all cells in the image from Cx22 dataset (the first row), where many cell fragments scatter irregularly in the image. When the cells overlap slightly with each other (the second row), our model can successfully segment all cytoplasm in the image. Also, the edges of cytoplasm masks predicted by our model are more accurate than those of Mask2Former. As for an EDF image (the third row), cells overlap at a very high rate, to make the edges of some cytoplasm hardly distinguishable. Figure 4d shows that some cells are not detected successfully by our model. Although our model could detect most of the cells in the testing dataset, approximately 20 percent of cells in images from ISBI2015 are mis-predicted. In summary, our proposed method performs better than the baseline model Mask2Former.

To further demonstrate the effectiveness of our model, a commonly used image from ISBI2014 dataset is selected as the input image to compare the cell segmentation results visually. For the sake of convenience, the segmentation results from literature²⁶ are used for comparison. As can be seen from Fig. 5, the edges of cytoplasm are much more precise than those of the compared methods. However, the image used for comparing the segmentation results of different methods has cells overlapping at a relatively low rate.

On the other hand, our previous researches have been conducted on cervical cytology image segmentation from traditional convolutional neural networks to transformer based models. We explored the cytoplasm segmentation on the two ISBI datasets by using Mask RCNN with PointRend module³⁸, to fine-tune the candidate edge pixels. Also, we delved into the characteristics of transformer structure and tackled the same segmentation problem using an enhanced DETR-based mask transformer³⁹. Another image from ISBI2014 dataset, where

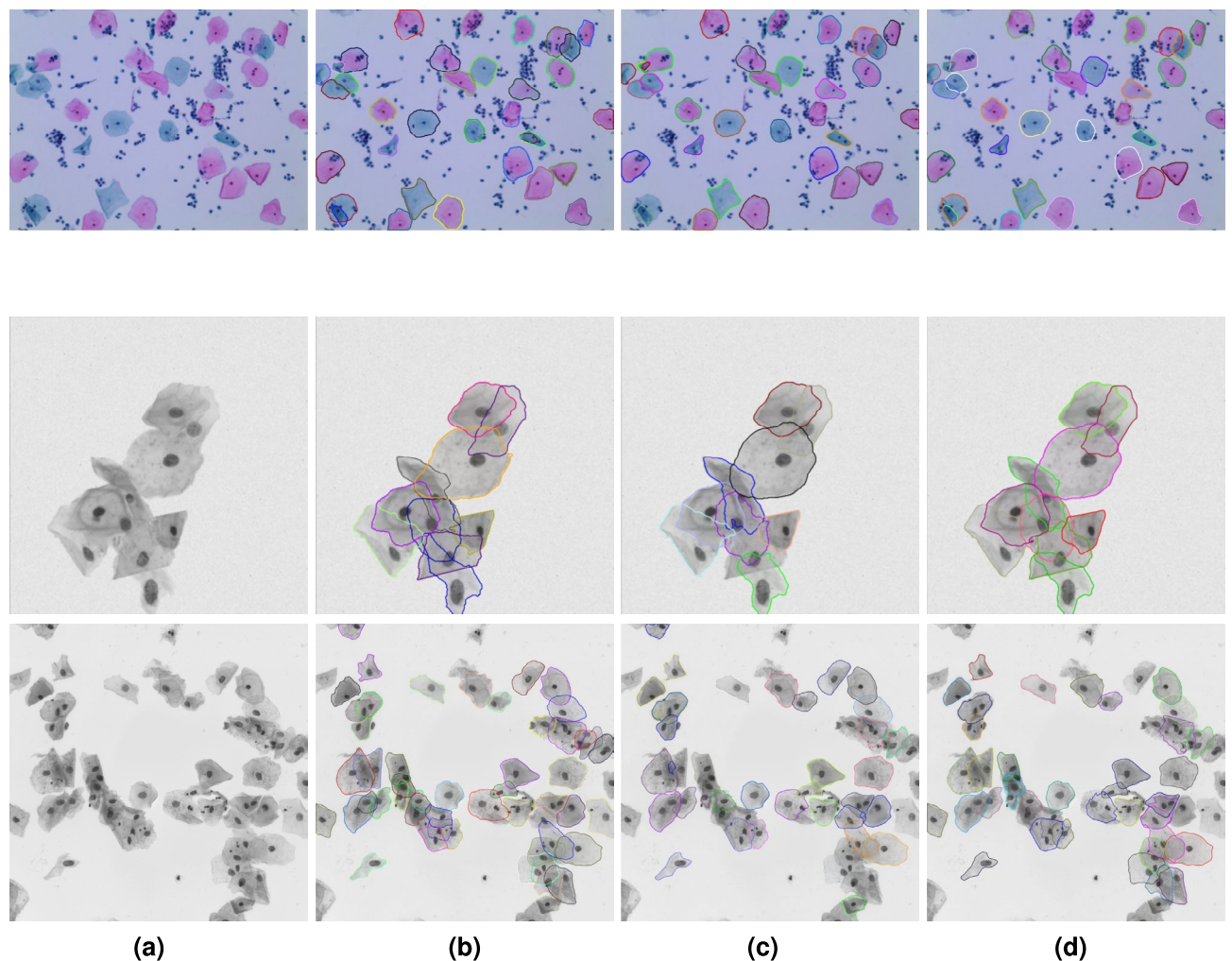


Figure 4. Three typical cytology images from the testing datasets (the first row from Cx22; the second row from ISBI2014; the third row from ISBI2015). The masks are predicted with DSC threshold=0.7. (a) The original input image for visual evaluation; (b) Ground truth segmentation of cytoplasm; (c) Cytoplasm segmentation predicted by the Mask2Former; (d) cytoplasm segmentation predicted by our model.

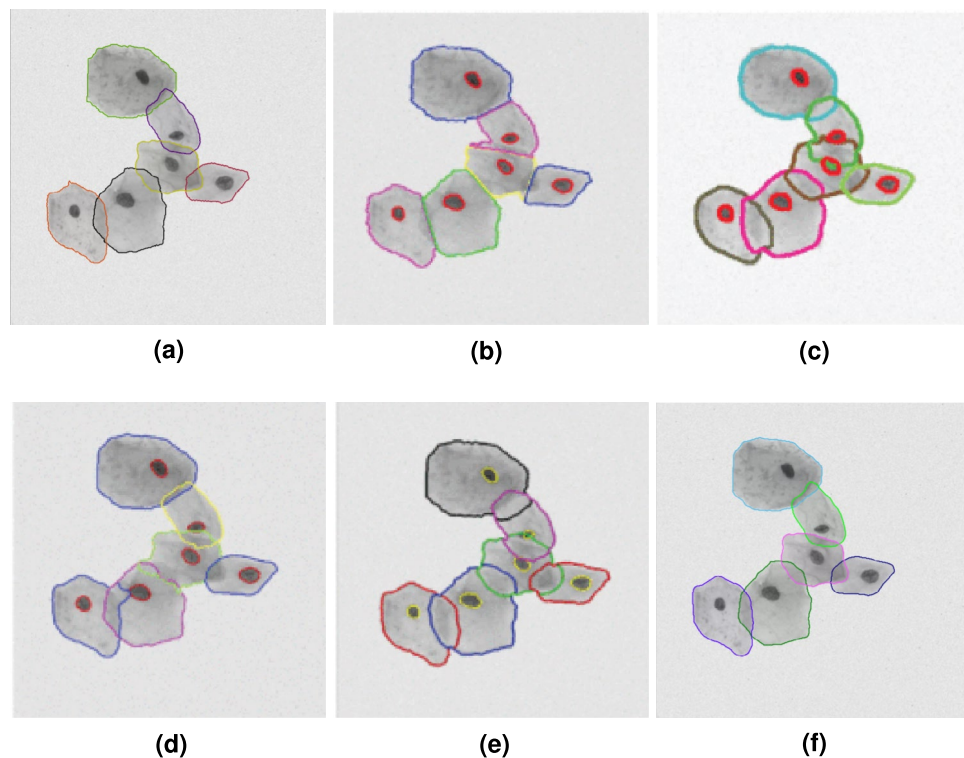


Figure 5. The qualitative segmentation results on an image from ISIB2014 dataset. (a) Ground-truth cytoplasm annotations. Images in (b), (c), (d), (e) are imported from Tareef et al.²⁶ (It should be noticed that those four images also contain the masks of nuclei). (f) Predicted cytoplasm segmentation by our method.

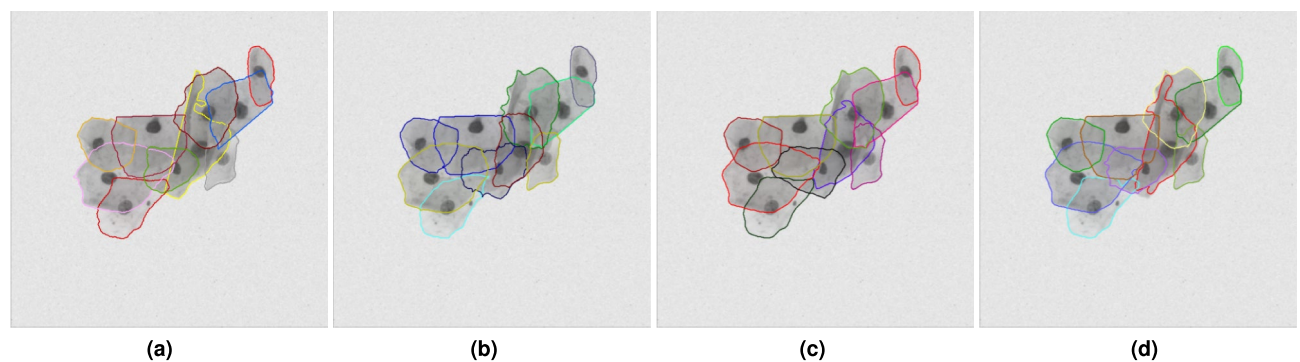


Figure 6. An image from ISIB2014 dataset with high overlapping rate is used to compare the segmentation results. The cytoplasm segmentation masks are predicted with DSC threshold=0.7. (a) The original input image with ground truth cytoplasm annotations; (b) The masks predicted by³⁸; (c) The masks predicted by³⁹; (d) The masks predicted by our model in this paper.

cells overlap higher than in Fig. 5, is selected to compare the segmentation performance of the proposed method with our previous works. As can be seen from Fig. 6, the masks predicted by our model in this paper are more accurate on the whole, compared with our previous models. For example, the cytoplasm mask with red edge in Fig. 6d is much closer to the ground truth mask (the one with yellow edge) in Fig. 6a than the masks in Fig. 6b (the one with brown edge) and Fig. 6c (the one with purple edge).

Discussion

Medical image segmentation has been proven to be very advantageous in aiding pathologists to analyze the conditions of patients. Recently Numerous deep learning based models, especially transformer-based networks, have been proposed to segment overlapping cervical cells. Transformer-based models typically use encoder module to acquire various semantic information in an image, and then the precise segmentation feature maps are obtained from the decoder module. Generally, networks of naive encoder-decoder structure tend to converge

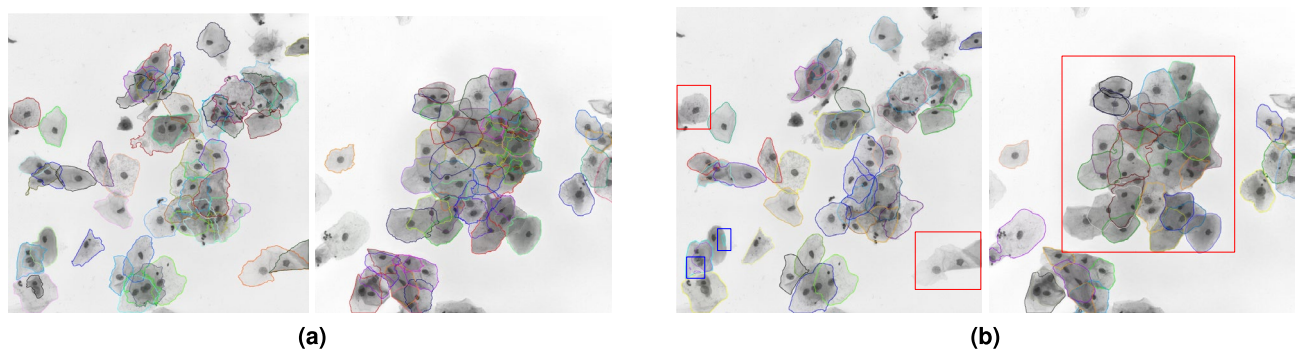


Figure 7. Two typical images from ISBI2015 dataset with very high overlapping rates and blurred edges. (a) The ground truth cytoplasm annotations; (b) The corresponding predicted masks by our model (with Swin_B as the backbone network).

slowly. In this paper, our approach aims to improve the segmentation performance on cervical cytology images and accelerate model convergence.

To evaluate the proposed model, numerous experiments were carried out using publicly available datasets, including ISBI and Cx22 datasets. As shown in Table 5, our model achieves state-of-the-art performance on all four metrics. Compared with the best result, the metrics DSC, TPRp and FNRo of our model obtain performance improvements by 3.4%, 2.3% and 2%, respectively. Particularly, the FNRo is 0, which means that every cytoplasm is successfully segmented by our model at the DSC threshold 0.7. However, Table 6 indicates that the performance of the proposed method is above the average level. As for the efficiency of training and inferring, it takes 18 hours for the model to run 300 epochs. And the inference for one image takes 0.208 second by average. Qualitatively, Fig. 4 shows that the edges of our model are more accurate than the baseline model Mask2Former on typical cytology images from three datasets. Figure 5 presents a visual comparison of segmentation results on a specific image from ISBI2014 between other outstanding models and our model.

Compared with our previous related works^{38,39}, the segmentation performance is equally matched. Some metrics in this paper are a little higher than previous methods and some metrics are a bit lower. Figure 6 shows that the edges of cytoplasm predicted by the model in this paper are a bit more accurate than the edges predicted by our previous models on an image from ISBI2014.

However, Table 6 shows that our model does not outperform other methods as significantly on ISBI2015 dataset as on ISBI2014 dataset (and also on Cx22 dataset). As can be seen in Figs. 1b and 4, the images in ISBI2015 dataset are far more complicated and blurred than the images in ISBI2014 (and Cx22) dataset. These characteristics of ISBI2015 dataset make a very challenging task even for skilled pathologists to identify the precise edges of all cytoplasm. On the other hand, there are only 8 images in the training dataset and 9 images in the testing dataset, which are somehow not sufficient for deep learning models. Neither the ResNet 50 nor the Swin_B backbone network obtains the features of the dense area effectively. And the content queries also fail to focus on the blurred cell edges in cell clumps. Figure 7 presents two typical images from the ISBI2015 dataset, which have two highest values of FNRo. In the left image of Fig. 7b, the cells in red rectangles are next to the border and not detected by our model; The predicted masks of cells in blue rectangles are not topologically equivalent to a circle, which should be suppressed. In the right image of Fig. 7b, the cells in the red rectangle are so dense that many cytoplasm are not detected by our model. The backbone network Swin_B uses a shifted window strategy to extract feature maps from original images. The features of cells adjacent to border may lack of surrounding pixel information, leading to misprediction of those cells. In short, a disadvantage of the proposed model is that it may fail to locate the cells in the most dense or margin areas.

To overcome the disadvantages of the proposed method, more effective structure for highly overlapped cells and constraints of edge curves are two possible ways. Firstly, special attention should be paid on improving the performance of segmenting cell instances from large cell masses. Under the hypothesis that a cell has only one nucleus and its corresponding cytoplasm, the detection of hardly overlapped nuclei may guide the segmentation of highly overlapped cytoplasm. Secondly, the ground truth annotations of cytoplasm are regions. The model is also trained to predict masks indicating possible cytoplasm regions. Then the dice loss could be calculated. Training the model to predict a curve (which is topologically equivalent to a circle) rather than a region may help the model focus on edge pixels.

Conclusion and future works

In this paper, we propose a novel transformer-based neural network to address the challenging task of segmenting cytoplasm from clumps of overlapping cells in cervical cytology images. We optimize the baseline model Mask2Former by adding class embeddings and noised ground truth masks, aiming to accelerate the convergence and improve the performance of the proposed model. The experiment results on publicly available datasets show that our approach outperforms state-of-the-art methods on both the ISBI2014 and the Cx22 datasets. And also the proposed model is slight better than the average level on the ISBI2015 dataset. Especially, the predicted masks on a randomly selected image (where cells overlap at a high rate) from ISBI2014 are more accurate than

the segmentation results of our previous works. Concretely, our proposed method (Swin_B) outperforms state-of-the-art result by 3.4% on DSC, 2.3% on TPRp and 2% on FNRo respectively, on the ISBI2014 dataset with DSC threshold = 0.7. The metric FNRo=20.2% on the ISBI2015 dataset is the lowest value ever. However, the metric FNRo on ISBI2015 dataset is still a little higher to some extent, compared with the corresponding value on ISBI2014 dataset. Our future work will focus on two possible ways to improve the performance of segmentation. One is that we will conduct in-depth research on the deep learning structure to find out more effective models for segmenting highly overlapped cells. The other is that the edge pixels of cell instances could serve as the targets to calculate the loss, rather than traditional dice loss. In other words, training the model to approach the edges rather than the regions may help improve the performance. Xia et al.⁴⁰ proposed a model ViT-CoMer, which facilitated bidirectional interaction between CNN and transformer. The model focused on dense predictions and therefore could serve as a possible improvement for our model.

Data availability

The two datasets analyzed during the current study were presented and made publicly available during the first and second Overlapping Cervical Image Segmentation Challenges at 2014 and 2015. The third dataset Cx22 was published by Liu et al. at 2022. ISBI2014: https://cs.adelaide.edu.au/~carneiro/isbi14_challenge/dataset.html ISBI2015: https://cs.adelaide.edu.au/~carneiro/isbi15_challenge/dataset.html Cx22: <https://github.com/LGQ330/Cx22>

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References

1. Siegel, R. L., Miller, K. D., Wagle, N. S. & Jemal, A. Cancer statistics. *CA Cancer J. Clin.* **73**(1), 17–48 (2023).
2. Sung, H. et al. Global cancer statistics 2020: Globocan estimates of incidence and mortality worldwide for 36 cancers in 185 countries. *CA Cancer J. Clin.* **71**(3), 209–249 (2021).
3. Carsen, S., & Marius, P. Cellpose3: one-click image restoration for improved cellular segmentation. *bioRxiv* 2024.02.10.579780. <https://doi.org/10.1101/2024.02.10.579780>. (2024).
4. Fu, Y. H., Liu, J. F. & Shi, J. TSCA-Net: Transformer based spatial-channel attention segmentation network for medical images. *Comput. Biol. Med.* **170**, 107938 (2024).
5. Zhang, B. C. et al. Cross-subject seizure detection in EEGs using deep transfer learning. *Comput. Math. Methods Med.* **2020**, 7902072. <https://doi.org/10.1155/2020/7902072> (2020).
6. Song, Y. Y. et al. Segmentation of overlapping cytoplasm in cervical smear images via adaptive shape priors extracted from contour fragments. *IEEE Trans. Med. Imaging* **38**(12), 2849–2862 (2019).
7. Lu, Z., Carneiro, G. & Bradley, A. P. An improved joint optimization of multiple level set functions for the segmentation of overlapping cervical cells. *IEEE Trans. Image Process.* **24**(4), 1261–1272 (2015).
8. Liu, J., et al. Local label point correction for edge detection of overlapping cervical cells, **2020**, [arXiv:2010.01919](https://arxiv.org/abs/2010.01919). (2020).
9. Lin, T. Y., et al. Microsoft COCO: common objects in context. Preprint at <https://doi.org/10.48550/arXiv.1405.0312>. (2014).
10. Zhou, B. L., Zhao, H., Puig, X., Fidler, S., Barriuso, A. & Torralba, A. Scene parsing through ADE20K dataset, *Computer Vision and Pattern Recognition (CVPR)*, (2017).
11. Ridnik, T., Ben-Baruch, E., Noy, A. & Zelnik-Manor, L. ImageNet-21K pretraining for the masses, **2021**, [arXiv:2104.10972v4](https://arxiv.org/abs/2104.10972v4). (2021).
12. Liu, G., Ding, Q., Luo, H., Sha, M., Li, X. & Ju, M. Cx22: A new publicly available dataset for deep learning-based segmentation of cervical cytology images, *Comput. Biol. Med.* **150**, Art No.106194. (2022).
13. Carion, N., Massa, F., Synnaeve, G., Usunier, N., Kirillov, A. & Zagoruyko, S. End-to-End Object Detection with Transformer, [arXiv:2005.12872v3](https://arxiv.org/abs/2005.12872v3). (2020).
14. Liu, S. L., et al. DAB-DETR: Dynamic anchor boxes are better queries for DETR, [arXiv:2201.12329](https://arxiv.org/abs/2201.12329), (2022).
15. Li, F., Zhang, H., Liu, S. L., Guo, J., Ni, L. M. & Zhang, L. DN-DETR: Accelerate DETR training by introducing query denoising, [arXiv:2203.01305v3](https://arxiv.org/abs/2203.01305v3), (2022).
16. Li, F., Zhang, H., Xu, H. Z., Liu, S. L., Zhang, L., Ni, L. M. & Shum, H. Y. Mask DINO: Towards a unified transformer-based framework for object detection and segmentation. [arXiv:2206.02777](https://arxiv.org/abs/2206.02777), (2022).
17. Liu, Z., et al., Swin Transformer: Hierarchical vision transformer using shifted windows, [arXiv:2103.14030](https://arxiv.org/abs/2103.14030). (2021).
18. Li, X. T. et al. Transformer-Based Visual Segmentation: A Survey. *IEEE Trans. Pattern Anal. Mach. Intell.* **46**(12), 10138–10163 (2024).
19. Cheng, B. W., Misra, I., Schwing, A. G., Kirillov, A. & Girdhar, R. Masked-attention mask transformer for universal image segmentation, [arXiv:2112.01523](https://arxiv.org/abs/2112.01523), (2022).
20. Ushizima, D. M., Bianci, A. G. C. & Carneiro, C. M. *Segmentation of subcellular compartments combining superpixel representation with Voronoi diagrams* (Lawrence Berkeley National Lab, Berkeley, CA, USA, 2015).
21. Norsrati, M. S. & Hamarneh, G. A variational approach for overlapping cell segmentation, In *Proc. IEEE ISBI Overlapping Cervical Cytology Image Segmentation Challenge*, 1–2(2–14), (2014).
22. Phoulady, H. A., Goldgof, D. & Hall, L. O. An approach for overlapping cell segmentation in multi-layer cervical cell volumes, The second overlapping cervical cytology image segmentation challenge-IEEE ISBI, (2015).
23. Phoulady, H. A., Goldgof, D., Hall, L. O. & Mouton, P. R. A framework for nucleus and overlapping cytoplasm segmentation in cervical cytology extended depth of field and volume image. *Comput. Med. Imaging Graphics* **59**, 38–49 (2017).
24. Lee, H. & Kim, J. Segmentation of overlapping cervical cells in microscopic images with superpixel partitioning and cell-wise contour refinement. In *2016 IEEE Conference on Computer Vision and Pattern Recognition Workshops (CVPRW)*, Las Vegas, NV, USA, 1367–1373(2016).
25. Wang, T., Lan, H., Niu, L. D., Fan, Z. P. & Yang, G. H. Overlapping cell segmentation of cervical cytology images based on nuclear radial boundary enhancement, *MOB INF SYST*, **2022**, Article ID 5969056, (2022).
26. Tareef, A. et al. Multi-pass fast watershed for accurate segmentation of overlapping cervical cells. *IEEE Trans. Med. Imaging* **37**(9), 2044–2059 (2018).
27. Tareef, A. et al. Optimizing the cervix cytological examination based on deep learning and dynamic shape modeling. *Neurocomputing* **248**, 28–40 (2017).
28. Wan, T., Xu, S. S., Sang, C., Jin, Y. L. & Qin, Z. C. Accurate segmentation of overlapping cells in cervical cytology with deep convolutional neural networks. *Neurocomputing* **365**, 157–170 (2019).
29. Zhang, H., Zhu, H. Q. & Ling, X. F. Polar coordinate sampling-based segmentation of overlapping cervical cells using attention U-Net and random walk. *Neurocomputing* **383**, 212–223 (2020).

30. Song, Y. Y. et al. Accurate cervical cell segmentation from overlapping clumps in Pap smear images. *IEEE Trans. Med. Imaging* **36**(1), 288–300 (2017).
31. Mahyari, T. L. & Dansereau, R. M. Multi-layer random walker image segmentation for overlapped cervical cells using probabilistic deep learning methods. *IET Image Process* **16**, 2959–2972 (2022).
32. Zhao, Y. L. et al. Automatic Segmentation of Cervical Cells Based on Star-Convex Polygons in Pap Smear Images. *Bioengineering* **10**(1), 47 (2022).
33. Jiang, H., Zhang, R. S., Zhou, Y. N., Wang, Y. M. & Chen, H. DoNet: Deep De-overlapping Network for Cytology Instance Segmentation, [arXiv:2303.14373](https://arxiv.org/abs/2303.14373), (2023).
34. Jiang, P., Liu, J., Feng, Jing., Chen, Hua., Chen, Y. Q., Li, C. & Pang, B. H. Interpretable detector for cervical cytology using self-attention and cell origin group guidance, *Engineering Applications of Artificial Intelligence*, **134** 108661 (2024).
35. Fang, M., Liao, B., Lei, X. J. & Wu, F. X. A systematic review on deep learning based methods for cervical images analysis. *Neurocomputing* **610**, 128630 (2024).
36. Chen, E., Ting, H. N., Chuah, J. H. & Zhao, J. Segmentation of overlapping cells in cervical cytology images: A survey. *IEEE Access* **12**, 114170–114189 (2024).
37. Huang, Q., Zhang, W., Chen, Y. L., Chen, J. Z. & Yang, Z. Review of cervical cell segmentation. *Multimed. Tools Appl.* <https://doi.org/10.1007/s11042-024-19799-0> (2024).
38. Zhang, B. C., Wang, W. F., Zhao, W., Jiang, X. L. & Patnaik, L. M. An improved approach for automated cervical cell segmentation with PointRend. *Sci. Rep.* **14**, 14210 (2024).
39. Zhang, B. C., Jiang, X. L. & Zhao, W. An enhanced mask transformer for overlapping cervical cell segmentation based on DETR. *IEEE Access* **12**, 176586–176597 (2024).
40. Xia, C. L., Wang, X. L., Lv, F., Hao, X. & Shi, Y. F. ViT-CoMer: Vision Transformer with Convolutional Multi-scale Feature Interaction for Dense Predictions. [arXiv: 2403.07392v3](https://arxiv.org/abs/2403.07392v3).

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Author contributions

Study design: B.Z., C.H. and W.L.; Segmentation experiments: B.Z, W.Z., and C.H.; Data analysis: B.Z., W.L and Q.L.; Preparing Tables and Figures: B.Z. and Q.L.; Writing original draft: B.Z. and W.Z.; Editing draft: C.H. and W.L.; All authors reviewed the manuscript.

Declarations

Competing interests

The authors declare no competing interests.

Additional information

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